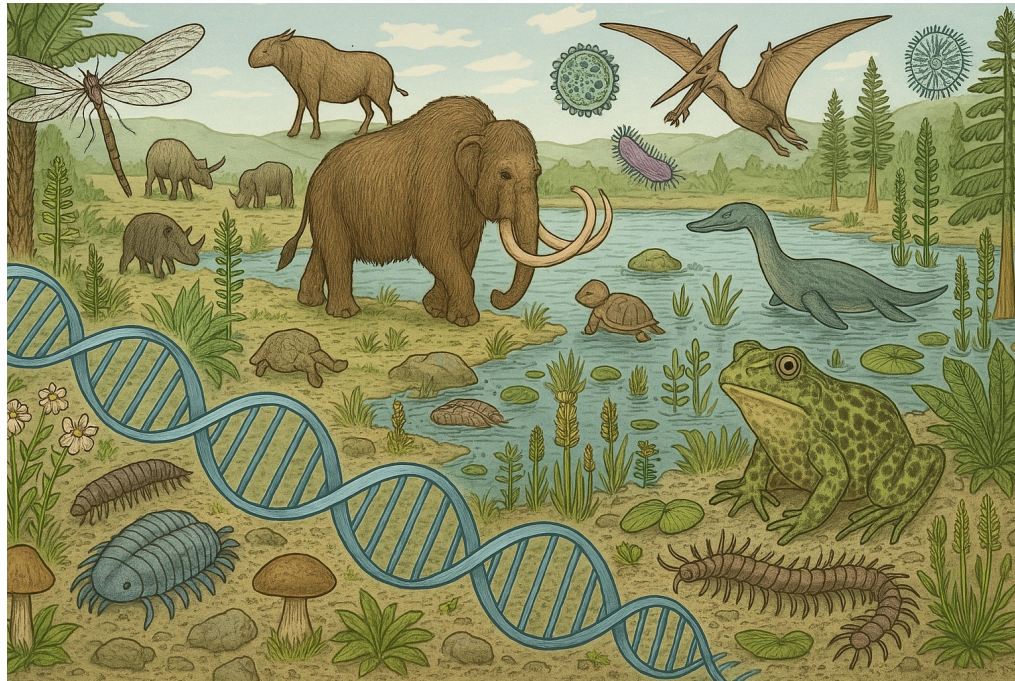


Ancient Metagenomics Analysis

Physalia course, online, 2-4 June 2026

Course outline and practical information

Nikolay Oskolkov, Metabolic Research Group leader, NIRI and Lund University



Physalia
Courses

About us

Organizer: Carlo Pecoraro, Physalia courses

info@physalia-courses.org

Instructor:

Dr. Nikolay Oskolkov, NIRI and Lund University

Nikolay.Oskolkov@biol.lu.se



@NikolayOskolkov



@osolkov.bsky.social



Personal homepage:
<https://nikolay-osolkov.com>

Brief introduction: who am I

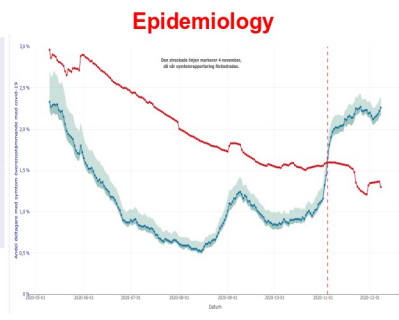
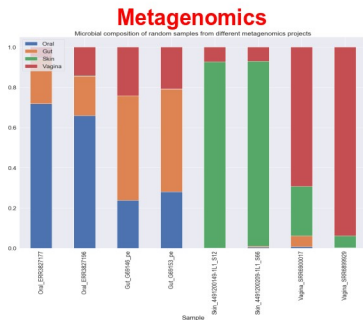
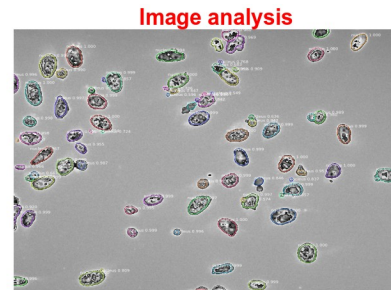
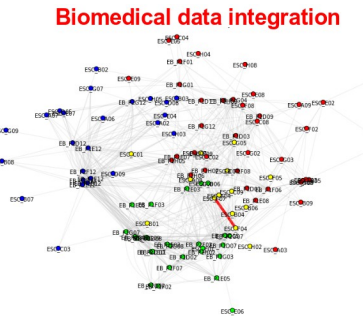
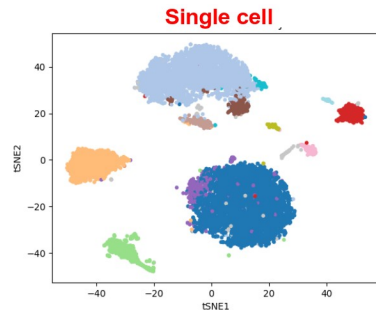
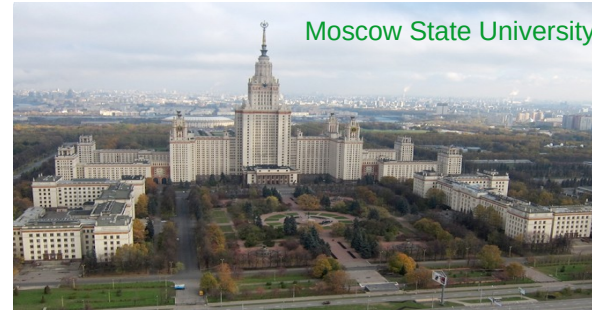
2007 PhD in theoretical physics in Moscow, Russia

2011 medical genomics at Lund University, Sweden

2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, LIOS, Latvia

60+ publications; h-index = 27; 4,300+ citations





Ph.D. Nikolay Oskolkov
Group Leader (PI) of the Metabolic
Research Group

Metabolic Research Group

The Metabolic Research Group (MRG) focuses on advancing computational methods to identify and validate novel drug targets for metabolic diseases. Our research profile centers on the development and application of machine learning approaches, combined with statistical modeling, to extract biological knowledge from complex datasets. A key expertise of the group is the integration of diverse multiOmics data—including genomics, transcriptomics, proteomics, metabolomics, and metagenomics—enabling a systems-level understanding of metabolic processes and disease mechanisms. Through this integrative and data-driven approach, we aim to contribute to precision medicine by supporting the discovery of innovative therapeutic strategies within the TARGETWISE project.



Kristina Grausa, PhD student
in Metabolic Research Group



Daniel Rivas, MD, PhD in AI,
postdoctoral fellow
in Metabolic Research Group

1 more postdoctoral fellow and 1 PhD student to be hired

**If you know anyone who might be interested,
please contact me!**

Publications

Conferences

[Metabolic Research Group](#)

METHOD

Open Access

aMeta: an accurate and memory-efficient ancient metagenomic profiling workflow



Zoé Pochon^{1,2†}, Nora Bergfeldt^{1,3,4†}, Emrah Kirdök⁵, Mário Vicente^{1,2}, Thijessen Naidoo^{1,2,6,7}, Tom van der Valk^{1,4}, N. Ezgi Altınışık⁸, Maja Krzewińska^{1,2}, Love Dalén^{1,3}, Anders Götherström^{1,2†}, Claudio Mirabello^{9†}, Per Unneberg^{10†} and Nikolay Oskolkov^{11†*}

[†]Zoé Pochon, Nora Bergfeldt, Anders Götherström, Claudio Mirabello, Per Unneberg, and Nikolay Oskolkov shared authorship.

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Full list of author information is available at the end of the article

Abstract

Analysis of microbial data from archaeological samples is a growing field with great potential for understanding ancient environments, lifestyles, and diseases. However, high error rates have been a challenge in ancient metagenomics, and the availability of computational frameworks that meet the demands of the field is limited. Here, we propose aMeta, an accurate metagenomic profiling workflow for ancient DNA designed to minimize the amount of false discoveries and computer memory requirements. Using simulated data, we benchmark aMeta against a current state-of-the-art workflow and demonstrate its superiority in microbial detection and authentication, as well as substantially lower usage of computer memory.

Keywords: Ancient metagenomics, Pathogen detection, Microbiome profiling, Ancient DNA

Background

Historically, ancient DNA (aDNA) studies have focused on human and faunal evolution and demography, extracting and analyzing predominantly eukaryotic aDNA [1–3]. With the development of next-generation sequencing (NGS) technologies, it was demonstrated that host-associated microbial aDNA from eukaryotic remains, which was previously treated as a sequencing by-product, can provide valuable information about ancient pandemics, lifestyle, and population migrations in the past [4–6]. Modern technologies have made it possible to study not only ancient microbiomes populating eukaryotic hosts, but also sedimentary ancient DNA (sedaDNA), which has rapidly become an independent branch of palaeogenetics, delivering unprecedented information about hominin and animal evolution without the need to analyze historical bones and teeth [7–12]. Previously available in microbial ecology, meta-barcoding methods lack validation and authentication power, and therefore, shotgun metagenomics has become the *de facto* standard in ancient microbiome research [13]. However, accurate detection,



aMeta: an accurate and memory-efficient ancient Metagenomic profiling workflow

snakemake Tests

About

aMeta is a Snakemake workflow for identifying microbial sequences in ancient DNA shotgun metagenomics samples. The workflow performs:

- trimming adapter sequences and removing reads shorter than 30 bp with Cutadapt
- quality control before and after trimming with FastQC and MultiQC
- taxonomic sequence kmer-based classification with KrakenUniq
- sequence alignment with Bowtie2 and screening for common microbial pathogens
- deamination pattern analysis with MapDamage2
- Lowest Common Ancestor (LCA) sequence alignment with Malt
- authentication and validation of identified microbial species with MaltExtract

When using aMeta and / or pre-built databases provided together with the workflow for your research projects, please cite our preprint: <https://www.biorxiv.org/content/10.1101/2022.10.03.510579v1>

Authors

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- Claudio Mirabello (@clami66) claudio.mirabello@scilifelab.se
- Per Unneberg (@percyfalc) per.unneberg@scilifelab.se

<https://github.com/NBISweden/aMeta>





GigaScience, 2025, 14, 1–14

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Technical Note

Improving taxonomic inference from ancient environmental metagenomes by masking microbial-like regions in reference genomes

Nikolay Oskolkov^{1,*}, Chenyu Jin^{2,3,4}, Samantha López Clinton^{2,3,4}, Benjamin Guinet^{2,3}, Flore Wijnands^{2,5}, Ernst Johnson^{2,5}, Verena E. Kutschera⁶, Cormac M. Kinsella^{3,7}, Peter D. Heintzman^{2,5}, and Tom van der Valk^{2,3,8}

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Abstract

Ancient environmental DNA is increasingly vital for reconstructing past ecosystems, particularly when paleontological and archaeological tissue remains are absent. Detecting ancient plant and animal DNA in environmental samples relies on using extensive eukaryotic reference genome databases for profiling metagenomics data. However, many eukaryotic genomes contain regions with high sequence similarity to microbial DNA, which can lead to the misclassification of bacterial and archaeal reads as eukaryotic. This issue is especially problematic in ancient eDNA datasets, where plant and animal DNA is typically present at very low abundance. In this study, we present a method for identifying bacterial- and archaeal-like sequences in eukaryotic genomes and apply it to nearly 3,000 reference genomes from NCBI RefSeq and GenBank (vertebrates, invertebrates, plants) as well as the 1,323 PhylorNorway plant genome assemblies from herbarium material from northern high-latitude regions. We find that microbial-like regions are widespread across eukaryotic genomes and provide a comprehensive resource of their genomic coordinates and taxonomic annotations. This resource enables the masking of microbial-like regions during profiling analyses, thereby improving the reliability of ancient environmental metagenomic datasets for downstream analyses.

Keywords: environmental DNA, ancient metagenomics, microbial-like regions

Introduction

Ancient environmental DNA (aeDNA) is a tool for studying past ecosystems, especially in contexts where traditional archaeological and paleontological tissue remains, such as bones and seeds, are absent [1–4]. It consists of genetic traces left by organisms in the environment, such as soil, sediments, or ice, and allows for the reconstruction of past biodiversity and ecological communities to provide insight into species extinction, vegetation changes, and ecosystem responses to climatic shifts and anthropogenic impacts.

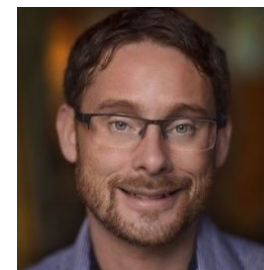
to genomic reference databases. Consequently, the quality of both the aeDNA data and the reference databases is crucial for reliable inferences. Microbial-like sequences in reference genomic databases, originating either from nonendogenous sources (contamination) or from evolutionary similarity to microbial genomes (e.g., due to ancient horizontal gene transfer or the endosymbiotic origins of plastids), can be a potential source of false-positive taxonomic identifications. In such cases, microbial sequences present in aeDNA data may be mistakenly classified as belonging to a eukaryotic reference genome due to sequence similarity.

NikolayOskolkov Update README.md			e45859c · 19 hours ago	48 Commits
data	modified nextflow pipeline		2 months ago	
images	Add files via upload		19 hours ago	
GTDB_fna2name.txt	added workflow files		7 months ago	
GTDB_sliced_seqs_sliding_window.fna.gz	added workflow files		7 months ago	
LICENSE.txt	Add files via upload		7 months ago	
README.md	Update README.md		19 hours ago	
detect_exogenous.sh	modified nextflow pipeline		2 months ago	
environment.yaml	added nextflow framework		2 months ago	
extract_coords.R	modified nextflow pipeline		2 months ago	
extract_coords_micr_contam.R	major modification of codes		2 months ago	
human_sliced_seqs_sliding_window.fna.gz	modified nextflow pipeline		2 months ago	
main.nf	modified nextflow pipeline		2 months ago	
micr_cont_detect.sh	major modification of codes		2 months ago	
nextflow.config	major modification of codes		2 months ago	
vignette.html	modified vignette		2 months ago	
vignette.jpynb	modified vignette		2 months ago	

GENEX
GENOME EXOGENOUS

GENome EXogenous (GENEX) sequence detection

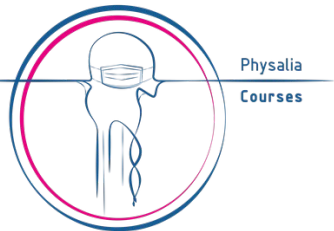
This is a computational workflow for detecting coordinates of microbial-like or human-like sequences in eukaryotic and prokaryotic reference genomes. The workflow accepts a reference genome in FASTA-format and outputs coordinates of microbial-like (human-like) regions in BED-format. The workflow builds a Bowtie2 index of the reference genome and aligns pre-computed microbial (GTDB v214 or NCBI RefSeq release 213) or



<https://github.com/NikolayOskolkov/MCWorkflow>

About you

- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course



Typical analysis methods used in ancient metagenomics

1) Alignment:



2) Classification:



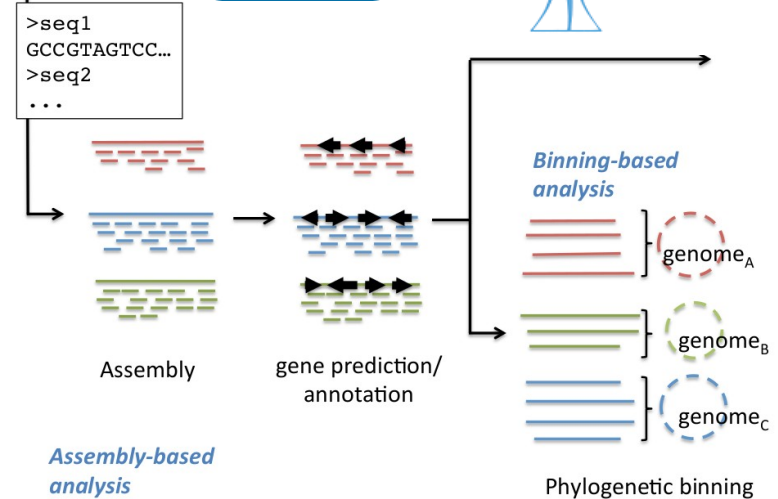
Centrifuge

MetaPhlan

Clark

Reference based:
assume similarity to reference

3) De-novo assembly:



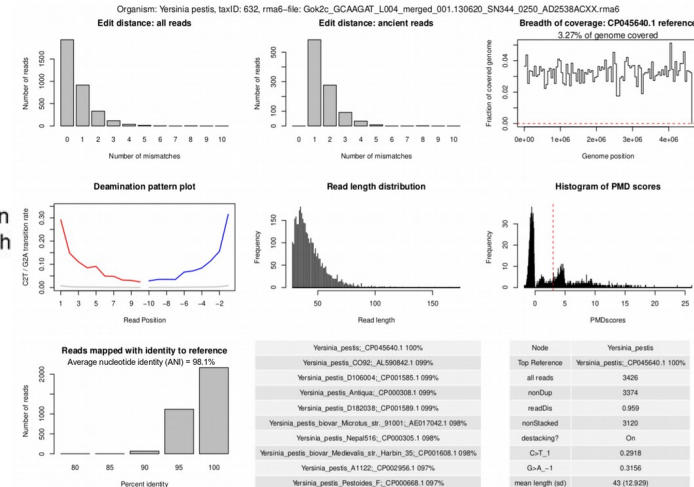
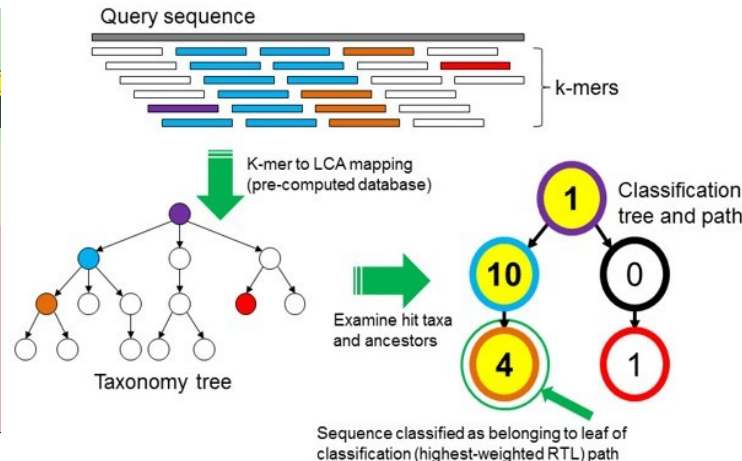
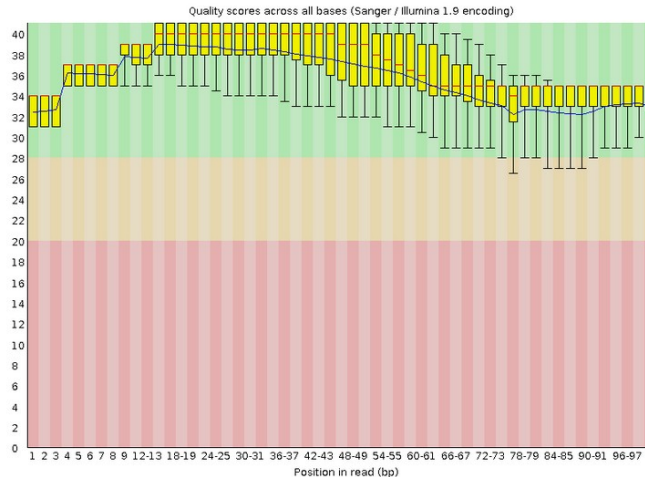
Reference free:
unbiased but challenging

Course outline

Day 1: introduction to ancient microbial and environmental metagenomics, quality control, adapter removal, host removal, taxonomic profiling

Day2: alignment, accessing mapping quality, decontamination, source tracking and authentication analyses, applying LCA to alignments for environmental samples

Day3: aMeta workflow, *de novo* assembly and authentication of assembled contigs



- To ask questions please **raise your hand, unmute yourself and ask**. You can also ask questions in the **zoom chat** or the **slack channel** for this course.
- I would greatly appreciate if you **keep your camera on** as much as possible for better contact and communication.
- The course includes **7 lectures (~1h each) and 7 practicals (~1.5h each)**, there is a **15 min break** after each lecture and each practical
- During practicals, we will use **bash command line tools**. At the end of each practical, I will go through command lines with my explanations.
- The course assumes **some familiarity with basics of bash / R / Python / aDNA**, you do not have to be fluent though. In case any terminology is unclear, please ask, I am ready to provide more detailed explanations.

Practical information: GitHub and Zoom

The course will take place in Zoom from 2 pm to 8 pm (CET, Berlin time)

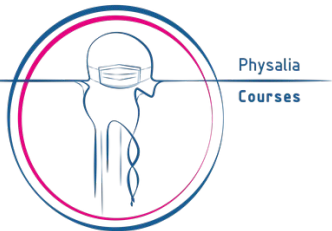
Links to the Zoom room will be posted in the Slack channel

The course GitHub repository containing lectures and exercises is:

<https://github.com/NikolayOskolkov/AncientMetagenomics>

Please bookmark this address! To copy the material to your computer, please do:

```
git clone https://github.com/NikolayOskolkov/AncientMetagenomics
```



Physalia_AncientMetagenomics_2025

Public

Pin

Unwatch 1

Fork 0

Star 0

master 1 Branch 0 Tags

Go to file

Add file

Code

LeandroRitter added articles	30370db · 7 hours ago	7 Commits
articles	added articles	7 hours ago
practicals	added material	11 hours ago
slides	added material	11 hours ago
README.md	added text to README	8 hours ago
command-line-basics.md	added material	11 hours ago
logo.png	changed logo	10 hours ago

README

About

No description, website, or topics provided.

Readme

Activity

0 stars

1 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages



Suggested workflows

Based on your tech stack

SLSA Generic generator

Generate SLSA3 provenance for your existing release workflows

Configure

Jekyll using Docker image

Package a Jekyll site using the jekyll/builder Docker image.

Configure

[More workflows](#)

Dismiss suggestions

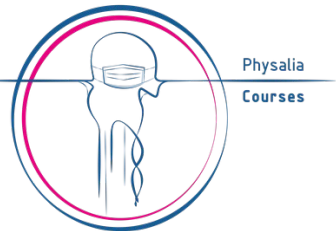
Practical information: Amazon Cloud (AWS EC2)



We will use the Cloud Computing service from Amazon, which we will access via `ssh` (secure shell protocol)

```
ssh -i ameta.pem -X ubuntu@54.244.40.220
```

- The IP address changes every day
- Everyone is given a username, with a `home` and `shared` folders
 - List of usernames can be found in Slack
 - The `shared` folder is copy-only: do not delete, move, rename, or write



To copy remote files and open them locally, please run:

```
scp -i ameta25.pem ubuntu@54.244.40.220:/remote_dir/remote_file /local_dir
```

ActivitiesTerminal

fre 20:31

sv🔊📄

nikolay@dell: ~/Documents/Consultation/Physalia/ancient_metagenomics

FileEditViewSearchTerminalHelp

(base) nikolay@dell:~/Documents/Consultation/Physalia/ancient_metagenomics\$ ssh -i ameta25.pem ubuntu@54.202.33.70

Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Thu May 22 08:18:12 UTC 2025

System load: 0.0

Processes: 165

Usage of /: 18.8% of 967.64GB

Users logged in: 0

Memory usage: 1%

IPv4 address for enX0: 172.31.36.154

Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

63 updates can be applied immediately.

To see these additional updates run: apt list --upgradable

8 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

Last login: Thu May 22 08:18:13 2025 from 130.235.244.193

(base) ubuntu@ip-172-31-36-154:~\$ ll

total 396508

drwxr-xr-x 14 ubuntu ubuntu4096 May 21 14:59 ./

drwxr-xr-x 40 root root4096 Jan 8 13:13 ../

-rw-r--r-- 1 ubuntu ubuntu61 Jan 27 11:34 .Xauthority

-rw-r--r-- 1 ubuntu ubuntu20865 May 21 16:25 .bash_history

-rw-r--r-- 1 ubuntu ubuntu220 Mar 31 2024 .bash_logout

-rw-r--r-- 1 ubuntu ubuntu4333 Jan 8 11:11 .bashrc

drwx----- 8 ubuntu ubuntu4096 May 21 12:52 .cache/

drwxrwxr-x 2 ubuntu ubuntu4096 Jan 9 12:13 .conda/

drwx----- 6 ubuntu ubuntu4096 Jan 28 14:35 .config/

drwxrwxr-x 4 ubuntu ubuntu4096 May 21 12:52 .java/

-rw-r--r-- 1 ubuntu ubuntu40 Jan 28 11:48 .lessht

drwxrwxr-x 3 ubuntu ubuntu4096 Jan 8 11:24 .local/

drwxrwxr-x 3 ubuntu ubuntu4096 May 21 12:51 .parallel/

-rw-r--r-- 1 ubuntu ubuntu807 Mar 31 2024 .profile

-rw-r--r-- 1 ubuntu ubuntu11 Jan 8 10:30 .python_history

drwx----- 2 ubuntu ubuntu4096 Jan 8 09:52 .ssh/

-rw-r--r-- 1 ubuntu ubuntu0 Jan 8 10:12 .sudo_as_admin_successful

-rw-r--r-- 1 ubuntu ubuntu16260 Jan 28 18:58 .viminfo

-rw-rw-r-- 1 ubuntu ubuntu381 May 21 15:28 .wget-hsts

-rw-rw-r-- 1 ubuntu ubuntu147784736 Jan 8 11:08 Miniconda3-latest-Linux-x86_64.sh

drwxr-xr-x 3 ubuntu ubuntu4096 Jan 8 13:26 R/

drwxr-xr-x 4 ubuntu ubuntu4096 May 21 15:04 Share/

drwxrwxr-x 12 ubuntu ubuntu4096 May 21 12:44 aMeta/

-rw-rw-r-- 1 ubuntu ubuntu1318204 Apr 21 2020 libssl1.1_1.1.1f-1ubuntu2_amd64.deb

drwxrwxr-x 19 ubuntu ubuntu4096 Jan 8 11:11 miniconda3/

-rw-r--r-- 1 ubuntu ubuntu574 Jan 13 10:57 r_packages_install.R

-rw-rw-r-- 1 ubuntu ubuntu128381702 Dec 16 19:50 rstudio-server-2024.12.0-467-amd64.deb

-rw-rw-r-- 1 ubuntu ubuntu128381702 Dec 16 19:50 rstudio-server-2024.12.0-467-amd64.deb.1

drwxrwxr-x 1 ubuntu ubuntu205 Jan 8 14:42 set_passwords.sh*